



THE UNIVERSITY OF LANCASTER

It is hereby certified that

Andrew Thomas

has been duly admitted to the degree of

Master of Engineering

Second Class Honours (Division One)

in

Electronic and Electrical Engineering

Andy J. Schofield.
Vice Chancellor

N. Mc...
Deputy Chief Executive
(Operations)

July, 2023

34669035-00167570-1





THE UNIVERSITY OF LANCASTER

Record of Learning and Achievement

34669035

Name of Student	Andrew Thomas	Date of Birth	01/07/2000
HESA Reference	1811231191675	Period of Study	01 Oct 2018 to 30 Jun 2023
Qualification	MEng Hons	Date of Award	10 Jul 2023
Scheme of Study	Electronic and Electrical Engineering		
Degree Classification	Upper second class		
Overall Aggregation Score	16.9		

Part 1 Course Modules Studied and the Grades Obtained

Course	Credit	Year	Title	Grade	Aggregation Score Out of 24
ENGRELC			Electronic Engineering		
ENGR113	8.0	18/19	Electrical and Electronics Fundamentals	I	17.5
ENGR114	8.0	18/19	Fundamentals of Electronic Instrumentation		
ENGR116	8.0	18/19	Fundamentals of Digital Electronics		
ENGR160	8.0	18/19	Process Engineering Fundamentals		
ENGR161	8.0	18/19	Fundamentals of Chemistry for Engineers		
ENGRMCH			Mechanical Engineering	IIi	14.6
ENGR111	8.0	18/19	Strength and materials		
ENGR112	8.0	18/19	Manufacturing Fundamentals		
ENGR117	8.0	18/19	Transport technology		
ENGR118	8.0	18/19	Heat Transfer		
ENGR119	8.0	18/19	Design, Innovation and 3-D Thinking		
MATH140			Mathematics for Engineers	I	21.3
ENGR115	8.0	18/19	Computers and Control		
MATH141	8.0	18/19	Determinants, Vectors and Further Complex Numbers		
MATH142	8.0	18/19	Elementary Calculus		
MATH143	8.0	18/19	Further Calculus and Numerical Techniques		
MATH144	8.0	18/19	Differential Equations		

Part II Course Modules Studied and the Grades Obtained

Course	Credit	Year	Title	Grade	Aggregation Score Out of 24
ENGR201	15.0	19/20	Engineering Analysis	I	21.9
ENGR202	15.0	19/20	Instrumentation and Control	IIi	17.4
ENGR203	15.0	19/20	Power and Heat	IIIi	16.4
ENGR204	15.0	19/20	Engineering Projects	IIii	14.0

Alison Mullan
ACADEMIC REGISTRAR

10/07/2023





THE UNIVERSITY OF LANCASTER

Record of Learning and Achievement

34669035

ENGR205	15.0	19/20	Business Development Project	III	9.0R
ENGR226	15.0	19/20	Electrical Circuits and Power Systems	III	16.4
ENGR227	15.0	19/20	Electromagnetics & RF Engineering	III	15.2
ENGR228	15.0	19/20	Digital Electronics	III	14.4
ENGR301	30.0	20/21	Individual MEng Project	III	17.4
ENGR311	15.0	20/21	Engineering Management	I	18.4
ENGR313	15.0	20/21	Power Electronics and Applications	I	18.2
ENGR332	15.0	20/21	Integrated Circuit Engineering	I	15.5
ENGR333	15.0	20/21	Analogue Electronics	III	12.9
ENGR335	15.0	20/21	Wireless Systems	I	21.0
ENGR336	15.0	20/21	Digital Signal Processing	III	16.3
ENGR445	15.0	20/21	Industry-linked Group Projects	I	17.9
ENGR400	0.0	22/23	Start-up Module	I	17.8
ENGR402	15.0	22/23	Advanced Embedded Systems	I	18.4
ENGR407	45.0	22/23	Group Project	I	19.9
ENGR408	15.0	22/23	Terahertz Science and Technology	III	14.5
ENGR424	15.0	22/23	Microengineering	I	19.3
ENGR427	15.0	22/23	High frequency electronics		

Alison Mullan

Alison Mullan
ACADEMIC REGISTRAR

10/07/2023



HIGHER EDUCATION ACHIEVEMENT REPORT (Diploma Supplement)

This Higher Education Achievement Report incorporates the model developed by the European Commission, Council of Europe and UNESCO/CEPES for the Diploma Supplement.

The purpose of the supplement is to provide sufficient recognition of qualifications (diplomas, degrees, certificates etc). It is designed to provide a description of the nature, level, context and status of the studies that were pursued and successfully completed by the individual named on the original qualifications to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why. In hard copy format this Higher Education Achievement Report is printed in black ink on paper watermarked with the crest of the University and carries the official University hologram. It is not valid unless in this format.

1 INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

- 1.1 Family name(s):
Thomas
- 1.2 Given name(s):
Andrew
- 1.3 Date of birth (day/month/year):
1 July 2000
- 1.4 Student identification number:
34669035

2 INFORMATION IDENTIFYING THE QUALIFICATION

- 2.1 Name of qualification and title conferred:
Master of Engineering with Honours
- 2.2 Main field(s) of study for the qualification:
Electronic and Electrical Engineering
- 2.3 Name and status of awarding institution:
Lancaster University
- 2.4 Name and status of delivering institution:
University of Lancaster
- 2.5 Language(s) of instruction/examination:
English

3 INFORMATION ON THE LEVEL OF THE QUALIFICATION

- 3.1 Level of qualification:
Integrated Masters (2nd cycle degree)
- 3.2 Official length of programme:
4 year(s) full-time study
- 3.3 Access requirements(s):
Detailed information regarding admission to the programme is available from the 'Find a course area' on our main web site at <http://www.lancs.ac.uk/>.

4 INFORMATION ON THE CONTENTS AND RESULTS GAINED

- 4.1 Mode of study:
Full Time
- 4.2 Programme requirements:
Please see next page(s)
- 4.3 Programme details:
Please see next page(s)
- 4.4 Grading scheme and, if available, grade distribution guidance:
Classification is based on the marks from years 2 and 3 (plus 4 where appropriate) and reflects the individual's performance across the whole degree programme. All marks are calculated on a 24 point scale. The calculation of the classification does vary slightly between programmes although the basic calculation is based on the students overall average aggregation score. Pass mark of 9
- 4.5 Overall classification of the qualification:
Upper second class
- 4.6 Date of Award:
10 July 2023

5 INFORMATION ON THE FUNCTION OF THE QUALIFICATION

- 5.1 Access to further study:
Access to postgraduate (2nd Cycle) study, normally if obtained with at least second class honours
- 5.2 Professional status (if applicable):
Not Applicable

6 ADDITIONAL INFORMATION

- 6.1 Additional information:
Please see next pages(s) if applicable
- 6.2 Further information sources:
www.lancaster.ac.uk





4.2 Programme requirements:

Graduates of this programme will:

Be able to write down and verbalise the fundamental concepts of Electronic and Electrical Engineering Science, complementing their otherwise general engineering background.

Be able to confidently select and apply appropriate mathematical methods and principles of Electronic and Electrical Engineering Science to model and analyse engineering situations.

Be able to write down and verbalise general principles of Electronic and Electrical Engineering design, design techniques and characteristics of engineering materials and components

Be able to apply Electronic and Electrical Engineering knowledge and understanding to the design a system, component or process to meet a need.

Be able to analyse existing engineering systems and create new systems through synthesis of ideas and Electronic and Electrical Engineering concepts from a range of sources.

Know the fundamentals of a range of manufacturing techniques and the requirements for their safe operation.

Be able to apply fundamental concepts of management and business practices to an engineering project.

Be able to undertake engineering projects cognisant of the professional and ethical responsibilities expected of the engineer.

Be able to work with others to manage a team project and be able to offer leadership in a team situation.

Demonstrate knowledge of technology-based leadership via team-based project activities and short industrial placement experiences.

Understand problems where knowledge can be incomplete and benefit from subsequent research and inquiry.

Be converse with and fulfil the engineering function expected of an Electronic and Electrical Engineer in a range of industrial contexts.

Fulfil the educational requirements of Chartered Engineer registration with the Institution of Engineering and Technology.

4.3 Programme details (e.g. modules or units studied), and the individual grades/marks/credits obtained:

Part I courses studied and passed:

Course	Year	Title	Aggregation Score	Percentage Mark	Credit	ECTS Credit
ENGRELC		Electronic Engineering				
ENGR113	18/19	Electrical and Electronics Fundamentals	17.5	68%		
ENGR114	18/19	Fundamentals of Electronic Instrumentation	14.1	57%	8.0	4
ENGR116	18/19	Fundamentals of Digital Electronics	15.0	60%	8.0	4
ENGR160	18/19	Process Engineering Fundamentals	21.1	81%	8.0	4
ENGR161	18/19	Fundamentals of Chemistry for Engineers	15.0	60%	8.0	4
ENGRMCH		Mechanical Engineering	22.2	88%	8.0	4
ENGR111	18/19	Strength and materials	14.6	59%		
ENGR112	18/19	Manufacturing Fundamentals	13.9	56%	8.0	4
ENGR117	18/19	Transport technology	13.3	54%	8.0	4
ENGR118	18/19	Heat Transfer	16.3	64%	8.0	4
ENGR119	18/19	Design, Innovation and 3-D Thinking	15.1	60%	8.0	4
MATH140		Mathematics for Engineers	14.3	58%	8.0	4
ENGR115	18/19	Computers and Control	21.3	82%		
MATH141	18/19	Determinants, Vectors and Further Complex Numbers	22.7	91%	8.0	4
MATH142	18/19	Elementary Calculus	22.0	87%	8.0	4
MATH143	18/19	Further Calculus and Numerical Techniques	21.5	83%	8.0	4
MATH144	18/19	Differential Equations	19.9	76%	8.0	4
			20.3	78%	8.0	4

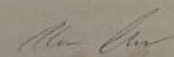
Part II courses studied and the grades obtained:

Course	Year	Title	Grade	Aggregation Score	Percentage Mark	Credit	ECTS Credit
ENGR201	19/20	Engineering Analysis	I	21.9	86%	15.0	8
ENGR202	19/20	Instrumentation and Control	Iii	17.4	68%	15.0	8
ENGR203	19/20	Power and Heat	Iii	16.4	65%	15.0	8
ENGR204	19/20	Engineering Projects	Iii	14.0	57%	15.0	8
ENGR205	19/20	Business Development Project	III	9.0 R	40%	15.0	8
ENGR226	19/20	Electrical Circuits and Power Systems	Iii	16.4	65%	15.0	8
ENGR227	19/20	Electromagnetics & RF Engineering	Iii	15.2	61%	15.0	8
ENGR228	19/20	Digital Electronics	Iii	14.4	58%	15.0	8
ENGR301	20/21	Individual MEng Project	Iii	17.4	68%	30.0	16
ENGR311	20/21	Engineering Management	I	18.4	71%	15.0	8
ENGR313	20/21	Power Electronics and Applications	I	18.2	71%	15.0	8
ENGR332	20/21	Integrated Circuit Engineering	Iii	15.5	62%	15.0	8
ENGR333	20/21	Analogue Electronics	Iii	12.9	53%	15.0	8
ENGR335	20/21	Wireless Systems	I	21.0	80%	15.0	8
ENGR336	20/21	Digital Signal Processing	Iii	16.3	64%	15.0	8
ENGR445	20/21	Industry-linked Group Projects	I	17.9	70%	15.0	8
ENGR400	22/23	Start-up Module				0.0	0
ENGR402	22/23	Advanced Embedded Systems	I	17.8	69%	15.0	8
ENGR407	22/23	Group Project	I	18.4	71%	45.0	22
ENGR408	22/23	Terahertz Science and Technology	I	19.9	76%	15.0	8
ENGR424	22/23	Microengineering	Iii	14.5	58%	15.0	8
ENGR427	22/23	High frequency electronics	I	19.3	74%	15.0	8
LANCAWARD	22/23	Lancaster Award				0.0	0

7 CERTIFICATION OF THE SUPPLEMENT

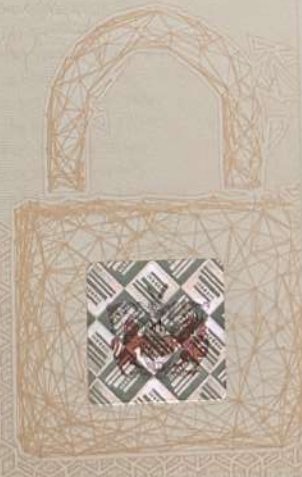
7.1 Date:
10 July 2023

7.2 Signature:


Alison Mullan

7.3 Capacity:
Academic Registrar

Andrew Thomas 34669035



8.2 Diagram of higher education qualification levels in England, Wales and Northern Ireland

In England, Wales and Northern Ireland¹, higher education institutions are independent, self-governing bodies active in teaching, research and scholarship. They are established by Royal Charter or legislation and most are part-funded by government.

Higher education (HE) is provided by many different types of institution. In addition to universities and university colleges, whose charters and statutes are made through the Privy Council which advises the Queen on the granting of Royal Charters and incorporation of universities, there are a number of publicly-designated and autonomous institutions within the higher education sector. Publicly funded higher education provision is available in some colleges of further education by the authority of another duly empowered institution. Teaching to prepare students for the award of higher education qualifications can be conducted in any higher education institution and in some further education colleges.

Degree awarding powers and the title 'university':

All universities and many higher education colleges have the legal power to develop their own courses and award their own degrees, as well as determine the conditions on which they are awarded. Some HE colleges and specialist institutions without these powers offer programmes, with varying extents of devolved authority, leading to the degrees of an institution which does have them. All universities in existence before 2005 have the power to award degrees on the basis of completion of taught courses and the power to award research degrees. From 2005, institutions in England and Wales that award only taught degrees ('first' and 'second cycle') and which meet certain numerical criteria, may also be permitted to use the title 'university'. Higher education institutions that award only taught degrees but which do not meet the numerical criteria may apply to use the title 'university college', although not all choose to do so. All of these institutions are subject to the same regulatory quality assurance and funding requirements as universities; and all institutions decide for themselves which students to admit and which staff to appoint. Degrees and other higher education qualifications are legally owned by the awarding institution, not by the state. The names of institutions with their own degree awarding powers ('Recognised Bodies') are available for download at:

<http://www.bis.gov.uk/policies/higher-education/recognised-uk-degrees/recognised-bodies>

Higher education institutions, further education colleges and other organisations able to offer courses leading to a degree of a Recognised Body are listed by the English, Welsh and Northern Irish authorities, and are known as "Listed Bodies". View the list at:

<http://www.bis.gov.uk/policies/higher-education/recognised-uk-degrees/listed-bodies>

Qualifications

The types of qualifications awarded by higher education institutions at sub-degree and undergraduate (first cycle) and postgraduate level (second and third cycles) are described in the Framework for Higher Education Qualifications in England, Wales and Northern Ireland (FHEQ). This also includes qualification descriptors that were developed with the HE sector by the Quality Assurance Agency for Higher Education (QAA - established in 1997 as an independent UK-wide body to monitor the standard of higher education provision - www.qaa.ac.uk). The FHEQ was self-certified as compatible with the Framework for Qualifications of the European Higher Education Area, the qualifications framework adopted as part of the Bologna Process, in February 2009. Foundation degrees, designed to create intermediate awards strongly oriented towards specific employment Higher Education Area they are "short

cycle" qualifications within the first cycle. The FHEQ is one component of the Credit and Qualifications Framework for Wales (CQFW). The Qualifications and Curriculum Authority (QCA), the Department for Children, Education, Lifelong Learning and Skills, Wales (DCELLS) and the Council for Curriculum Examination and Assessment, Northern Ireland (CCEA) have established the Qualifications and Credit Framework (to replace, in time, the National Qualifications Framework (NQF)). These authorities regulate a number of professional, statutory and other awarding bodies which control VET and general qualifications at all levels. The QCF is also incorporated into the CQFW. There is a close association between the levels of the FHEQ and the NQF (as shown overleaf), and other frameworks of the UK and Ireland (see 'Qualifications can cross Boundaries'.

<http://www.qaa.ac.uk/Publications/InformationAndGuidance/Pages/qualifications-can-cross-boundaries.aspx>

Quality Assurance

Academic standards are established and maintained by higher education institutions themselves using an extensive and sophisticated range of shared quality assurance approaches and structures. Standards and quality in institutions are underpinned by the universal use of external examiners, a standard set of indicators and other reports, by the activities of the QAA, and in professional areas by relevant professional, statutory and regulatory bodies. This ensures that institutions meet national expectations described in the FHEQ: subject benchmark statements, the Code of Practice and programme specifications. QAA conducts peer-review based audits and reviews of higher education institutions with the opportunity for subject-based review as the need arises. The accuracy and adequacy of quality-related information published by the higher education institutions is also reviewed. QAA also reviews publicly funded higher education provision in further education colleges.

Credit Systems

Most higher education institutions in England and Northern Ireland belong to one of several credit consortia and some operate local credit accumulation and transfer systems for students moving between programmes and/or institutions. A framework of national guidelines, the Higher Education Credit Framework for England, was launched in 2008. Credit is also an integral part of the CQFW and the QCF. It may be possible for credit awarded in one framework to be recognised by education providers whose qualifications sit within a different framework. HE credit systems in use in England, Wales and Northern Ireland are compatible with the European Credit Transfer System (ECTS) for accumulation and transfers within the European Higher Education Area, and are used to recognise learning gained by students in institutions elsewhere in Europe.

Admission

The most common qualification for entry to higher education is the General Certificate of Education at 'Advanced' (A) level. Other appropriate NQF level 3 qualifications and the kite-marked Access to HE Diploma may also provide entry to HE. Level 3 qualifications in the CQFW, including the Welsh Baccalaureate, also provide entry, as do Scottish Highers, Advanced Highers or qualifications at the same levels of the Scottish Credit and Qualifications Framework. Part-time and mature students may enter HE with these qualifications or alternatives with evidenced equivalent prior formal and/or experiential learning. Institutions will admit students whom they believe to have the potential to complete their programmes successfully.

¹ The UK has a system of devolved government, including for higher education, to Scotland, to Wales and to Northern Ireland. This description is approved by the High Level Policy Forum which includes representatives of the Department for Education and Skills (DfES), Scottish Executive, the Welsh Assembly Government, the Higher Education Funding Councils for England, Scotland and Wales, the Quality Assurance Agency for Higher Education (QAA), the Standing Conference of Principals and the National Recognition Information Centre for the UK (UK NARIC).



8.2 Diagram of higher education qualification levels in England, Wales and Northern Ireland

